Security Noir: Critical Designs for Security & Privacy

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In this paper, we investigate an alternative method for communicating the somewhat elusive security risks and privacy invasions of technology, the web, social networks, and other aspects of modern life. Specifically, we draw from the *critical design* movement popularized by Dunne and Raby in the 90s. In this project, we have conceptualized 10 new technological designs and described their use through short vignettes. These designs are imaginary but plausible technologies that illustrate a security or privacy risk through absurdity, satire, or humour. We also consider the ability to co-design with generative artificial intelligence tools, finding the tools cannot (yet?) replicate our design process and produce satisfactory designs, but these tools are useful today in a 'co-pilot' role for generating high volumes of ideas for human designers to curate and refine.

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1 Introductory Remarks

(LEAH: MAYBE WE CAN OPEN WITH ONE OF OUR VIGNETTES AS A HOOK LIKE THIS? I ALSO ADDED A BIT MORE CONTEXT FOR THE INTRO)

Consider the following vignette from a collection of speculative artefacts:

> **Eleventh Finger.** Biometric authentication based on fingerprints is generally user-friendly and fast. Eleventh finger is a 3D printed rubber finger, customized with the user's fingerprint. You can put it on a keychain and use it on cold winter days when you do not want to remove your gloves. Alice lets her friend Bob borrow it when he stays at her house. It can also serve as a backup if the worst happens.

At face value, the design is a convenient-enabling accessory for biometric authentication; yet, the vignette also challenges common assumptions about fingerprints as secure, embodied identifiers. But what happens when bodily identity is lent, misplaced, duplicated, or stolen? The scenario does not present a technical solution; instead, it invites reflection.

This mode of engagement is characteristic of critical design, which involves speculative artifacts and narratives that do not propose deployable products, but rather to provoke reflection and discussion. Critical design is commonly attributed to Dunne and Raby [5, 7] however it can be seen as a refinement of earlier movements in arts and design, beginning with the Italian radical design movement and extending through critical practice in HCI [20]. Consider the three examples provided in Table 1. Each describes the design of an object or technology. In contrast to industrial design, these technologies are not oriented around problem-solving, nor the creation of solutions that are functional, useful, and pleasurable, nor are they meant for mass production. Rather they offer a subtle criticism by illustrating something—electromagnetic radiation, the inevitability of death, the ubiquity and stealth of surveillance—that typically

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Table 1. An Example of Critical Design

Work	Vignette
Compass Table, A. Dunne F. Raby	"This table reminds you that electronic objects extend beyond their visible limits. The 25 compasses set into its surface twitch and spin when objects like mobile phones or laptop computers are placed on it. The twitching needles can be interpreted as being either sinister or charming, depending on the viewer's state of mind. When we designed the compass table, we wondered if a neat-freak might try to make all the needles line up, ignoring the architectural space of the room in favour of the Earth's magnetic field." Quoted from [7].

goes unquestioned. They are designed to invoke a dilemma for the user, raising questions, challenging values, and stimulating debate. Critical design considers these goals to be the function (or 'para-function' [5]) of the concept.

Critical design has significant overlap with conceptual art, utilizing common methods, and has been disseminated through exhibitions and the art literature. However critical design works by showcasing things that are plausible, believable, and easy for the reader to situate, requiring design practice. Viewing critical design as art creates challenges. Consider the compass table (from Table 1) in the context of an art gallery exhibit—it is meant to illustrate electromagnatic radition in every day life, in one's home, in different positions and orientations within that home—all of this is lost when it sits statically in a gallery. Should critical design be displayed in galleries, a vignette or film or narrative will often accompany the object to describe its context of use? Security and privacy risks are often intangible and invisible, making concepts difficult to communicate to everyday users. We saw a parallel to electromagnetic radiation in this regard, and considered if critical design could similarly aid in illuminating risks.

In this Provocations paper, we explore alternative methods of communicating elusive security and privacy risks through critical designs. We present ten speculative technological creation through short vignettes that leverage humour, satire, and exaggeration to render otherwise intangible concerns perceptible and reflectable. While the designs are exploratory, their grounding aligns with perspectives from uable privacy and security research that emphasize how users construct understandings of risk informal channels, including entertainment media [CITATION] and interpersonal stories [CITATION], play a significant role in shaping users' awareness and interpretation of privacy and security risks. By presenting plausible but exaggerated artefacts embedded in narrative contexts, critical designs function as communicative probes that externalize otherwise invisible socio-technical tensions. We argue that if users already derive security lessons from mediated narratives, then critically constructed artefacts may serve a similar pedagogical role by acting as designed media that foreground overlooked vulnerabilities and provoke reconsideration of taken-for-granted technological assumptions. From this perspective, critical design is not merely artistic critique, but could function as a viable exploratory tool for risk communication.

1.1 Related Work

Critical Design for Security and Privacy. Much has been written on critical design and Malpass provides an excellent survey [20]. Since our focus is security and privacy, we only summarize the following concepts, situated within or close to critical design, since they deal with issues of privacy:

- **The Pillow.** Dunne and Gaver's design takes the form factor of a transparent pillow and plays radio signals it picks up from the technology of that time: *e.g.*, phones, pagers, and baby monitors [6]. It challenges users to consider how confidential these signals are.

- **Open Informant.** Described in Table 1, this design exposes nation-state surveillance.
- **United Mirco-Kingdoms.** Dunne and Raby reimagine the UK as four 'supershires' where one is occupied by 'digitarians:' extreme technocrats ruled by algorithms in a state of total surveillance.
- **Alternet.** While not critical design, Gold's Alternet has the similar intention of provoking thought through design [11]. The internet is re-imagined with users retaining full control over their private data, touching on web tracking and our growing digital dossiers.

We are unaware of any attempt to build a systematic set of examples based on security or privacy, nor much attention to it from academic research papers, however critical design has been used in other areas of computer science. The uptake of critical design in HCI research is encouraged based on the anticipated value critical design will add to the field [1]. Speculative design is another related concept that HCI researchers rely on. An example is to analyze and question the usage of a certain technology and the consequences of it in the near future [16]. Ludic design is also utilised to create designs that do not dictate a certain purpose or usage but rather leave the interpretation to users [10].

Methodologies for Design Creation and Evaluation. While design process can be based on well-established methods, that are relevant to us, like whiteboarding, using sticky notes, affinity diagrams; the integration of AI to the creation process has been proliferated with the abundance and the variety of AI-based tools. These tools can be used to generate the content itself or help the designer along the process of creation and ideation. There are also works that examine how users prompt generative AI ([26], [3]); utilize generative AI for co-ideation or co-creation [28],[4].

There are also efforts towards creation of computational tools to that would assist task that are manually done in general. One line of work is about creating tools to group design aspects to help ideation and exploration of design solutions[15]. Speculative design is also used to imagine the future of the tools based on generative AI [18]. Custom-made AI tools are proposed for specific design goals like integrating context awareness[8].

While we only created the designs and did not apply any techniques to evaluate them, there are several methodologies for this purpose. User studies like interviews and surveys are one of the most common techniques employed [26], [3]. They are deployed to evaluate different aspects of the design process like assesing the utility of the AI-tools used during the design process, understanding users' needs or deploying a participatory workshop. Methods to asses the creativity of AI-generated content are also proposed [2].

Alternative Pedagogy. Somewhat related to critical design, there are works on finding alternative ways (like design fiction [19], role playing game [21], science fiction and design fiction [29]) to explain tricky concepts not only in security but also in other fields like sensing technologies.

CS education literature has explored how to teach concepts related to computer science in general or cybersecurity to a diverse audience. Some of which are understanding how CS students perceive AI and cybersecurity [24], using visual cryptography for explaining concepts to K-12 students [25], how to initiate a security mindset [14]. Studying the metaphors used in CS education gives important insights about how students perceive certain concepts [13].

2 Methodology A: Traditional Design

2.1 Methodology

Our research goal was to create critical design vignettes illustrating security and privacy issues. We will first describe our design methodology for human designers. We began by reviewing the critical design literature, and we settled on delivering vignettes, which would be short paragraphs describing a new technology and illustrating, with short

narratives, how it functions and the feelings it might invoke. The technology is meant to be plausible in the near future (the reader can conceive why someone, somewhere might think it is a good idea), while also showing the darker side of convenience, risks that are normalized and overlooked, or ways a user might be betrayed by the technology. During this period, we were encouraged to notice 'found examples' of critical design in the wild, although perhaps not expressed that way.[1]

We settled, after some trial and error, on a digital (Apple Freeform) 'sticky notes' methodology to generate affinity maps, where we clustered our notes in four groups: (1) security and privacy concerns (*e.g.,* 'if you don't pay for the product, you are the product'), misconceptions (*e.g.,* 'nothing to hide'), or principles (*e.g.,* 'minimal disclosure') that could be the basis of a critique; (2) domains where users might especially care about security or privacy (*e.g.,* health records or social networks); (3) a form factor for the new technology (*e.g.,* haptics or extended reality); and (4) a possible user response (*e.g.,* amused or embarrassed). Figure **??** shows our board. Artefacts, including this board, are available on our GitHub repository.[2] We brainstormed around individual notes and drew random combinations to spur divergent thought. The design methodology provided structure but did not make designing simple. Each of our 10 designs was quite difficult to develop, each taking many hours of discussion and dozens of discarded ideas that we could not translate into designs.

2.2 Designs

Crystal Avatars. Crystal avatars are created instantly but only gets better over time. At first, they are not recognizable and cannot be differentiated from other users, but as users browse the site, small details crystallize. Alice is happy to save time registering for the site and have a personal avatar, which automatically added a cat accessory after she spent a lunch hour reading about hairball treatments. However she grows uneasy about the "Woman, Life, Freedom" badge, a cause she believes in but only vocalizes discriminately.

[1]For example, well-known cryptographer Matthew Green tweeted: *There's an entire class of browsing many folks will feel uncomfortable doing if their browser has a little icon in the window that says "Hi <User> I know this is you!" This is Human Behavior 101.*
[2]GitHub: https://github.com/MadibaGroup/2017-Critical-Design

Calm Watch. Calm Watch is a smart watch that tracks anxiety levels based on several biomarkers. When the anxiety level of the wearer crosses a threshold, a haptic vibration pulse prompts the user to look at the watch. From there, they can select a variety of calming techniques, including breathing exercises and guided meditation. Bob is nervous to talk to Alice and becomes flustered when his watch starts vibrating mid-conversation, just loud enough for Alice to hear it and notice his fidgeting with the watch.

Chatty. The latest software update for Alice's voice-activated home assistant Chatty adds machine learning to better infer Alice's commands, such as adding items to her shopping list and suggesting music she might like. It does not always know whether Alice is talking to it or not, so over time it picks up on pieces of Alice's life and has trouble unlearning them. Alice is startled but pleased when Chatty chimes in with the name of 'that actor who was in that thing' she was telling her friend about. She is bemused to find a specialized vinyl cleaning solution on her shopping list after she said Bob's records smells fishy.

Dynamic Laughtrack. Digital television content, such as sitcoms, encode laughtracks as a series of cues rather than an actual recording. Smart TVs listen and classify the viewer's level of laughter on a 10 point scale (with fine grain training over time). The TV adds laughter to the content in a sensitive and considerate manner, where the laughtrack is only marginally higher on the scale than the viewer's own laughter. This gently nudges the viewer to greater enjoyment of the program without bombarding her. It also mixes in actual past recordings of the viewer's own laughter to capitalize on emotional mirroring.

Eleventh Finger. Biometric authentication based on fingerprints is generally user-friendly and fast. Eleventh finger is a 3D printed rubber finger, customized with the user's fingerprint. You can put it on a keychain and use it on cold winter days when you do not want to remove your gloves. Alice lets her friend Bob borrow it when he stays at her house. It can also serve as a backup if the worst happens.

GOTTCHA. GOTTCHA is a human-detection system to protect online accounts being accessed by bots. It invokes the device camera to analyze if a live human is using the device. To protect against video replays or machine-generated video, it unexpectedly prompts the user with a randomly selected image and carefully measures their reaction to it. Micro-expressions are involuntary facial displays of emotion that are too fast to mimic artificially. GOTTCHA's image bank can provoke disgust, anger, fear, sadness, happiness, surprise or contempt. Over time, Alice stops visiting websites that use GOTTCHA because of its capricious tendency to mix in disturbing images.

No(i)sy elevator. Elevator music has never been regarded as elevating your mood. This is why employees will be happier riding the noisy elevator. When it recognizes a user, it spins up a selection from their most-played songs on major music streaming platforms. Alice finds hearing her favourite Miles Davis song grounds her at the start of each work day, while Bob and Carol, coincidently riding the elevator together, discover their shared love for 90s britpop. David is a bit more concerned about the profanity ridden banger that the nosy elevator plays for him and his boss.

Receiver plant. The receiver plant in the front yard of Alice records the information of the people that pass by. It is a greedy plant that needs to absorb enough bluetooth data to grow. Bob is singlehandedly responsible for its lustrous canopy from walking by Alice's house every day, mostly while checking his work emails on his phone. Perhaps the plant overstepped when it displayed, "you are 15 minutes late today, are you sure you can make it to work on time?"

ToS Fishing. ToS Fishing is a digital game where players earn points by hunting down excessive legalese on websites, including terms of service (ToS), privacy policies, and cookie policies. To play, users simply copy and paste the URL of the legalese. If the ToS has been seen before, the users are awarded points based on its word count. If it has not been seen before, it is reviewed by a human for validity and word count—and in this case, the user gets a finder's bonus. To mitigate people from launching custom websites, the game only accepts sites from the Alexis top million. Users display their aggregate score on a leaderboard with a profile showing their top catches (word count only, not the website), and can earn various badges for playing consistently.

Wifi Projector. Minimalist wifi routers have a single light indicator to show internet connectivity, such as a green light as opposed to red light. This understates the interesting and intriguing data flowing through the device as users browse the internet. By contrast, the wifi projector is a talkative router that projects visualizations onto the ceiling above where it is placed. Every website that pulls in scripts and cookies from other domains is displayed as a constellation of stars of various colours and sizes, which slowly fade away. While Alice's visit to dictionary.com is a beautiful galaxy, she is also perturbed by the number of ads and tracking companies surveilling her.

2.3 Discussion

(LEAH: PERHAPS THESE COULD BE CONVERTED INTO DISCUSION POINTS, LIKE HOW WE EXPECT PEOPLE TO USE THE VIGNETTES, AND OUR THOUGHTS ON THE METHODOLOGY TO DEVELOP THEM? I ADDED A BIT OF TEXT ABOUT VIGNETTE AND THEIR USE IN HCI AND USABLE PRIVACY AND SECURITY)

Vignettes. Critical designs are commonly described through vignettes. Occasionally they are built and displayed. In the case of Dunne and Raby's Compass Table (and the seven other objects in their *Placebo* project [7]), the table was built and given to individuals to use. This field study was followed by an exit survey on how the participants felt about the object. In this paper, we describe our designs through short vignettes that illustrate its design and use case. If they are a successful design, the reader should easily imagine its use in their life and question it critically, resulting in its 'rhetorical use' [20] rather than practical use. These designs are transgressive and intended to provoke the reader's emotions; invoking humour, satire, ridicule, poetry, playfulness, lewdness, appropriation, irreverence, or deliberate overcompensation.

Vignettes, defined as "short stories about hypothetical characters in specified circumstances, to whose situation the interviewee is invited to respond" [9] are also established instruments in speculative design, design fiction, and scenario-based inquiry in HCI. In these contexts, short narratives depictions of hypothetical technologies or practices are used to surface assumptions, explore value tensions, and prompt reflection on possible futures without requiring the artefact itself to exist materially. Design fiction work in HCI, in particular, are not treated as empirical stimuli for behavioural measurement, but as structured prompts that enable participants to reason about situated technological futures, their consequences, tradeoffs, and social norms. Vignette-based methods in usable security research (e.g., [22], [12]) similarily construct narratives to investigate user expectations, decision-making processes, and ethical considerations surrounding emerging technologies when direct obervation or deployment is infeasive, unsafe, or premature.

In this work, we propose vignettes as a methodological vehicle for communicating speculative artefacts rather than as experimental instruments intended to elicit measurable responses. The vignettes function as analytic artefacts: they instantiate design concepts in situated contexts, articulate their purpose and functionality, and embed a short narrative

through which readers can interpret and reflect on the security and privacy tensions being surfaced. This approach aligns with established HCI practices in which narrative scenarios are used to probe socio-technical implications. However, it does not aim to produce generalizable behavioural data or predictive claims about user action. Instead, our objective is interpretive and exploratory — to expose assumptions, provoke critical reasoning, and broaden discourse around the topics under investigation.

Beyond their role within this paper, these vignettes are intended to be shareable artefacts that others may appropriate for reflection, inquiry, and design exploration. While we invite open interpretation in their application, we anticipate several avenues of use. First, critical design vignettes may serve as reflective prompts in educational, participatory, or professional contexts, where speculative scenarios facilitate discussion of assumptions, values, and mental models surrounding security and privacy practices. Second, they may function as research instruments in future speculative design or vignette-based studies, where they can be adapted as stimuli to explore perceptions of risk, trust, or technological futures. Finally, we envision their use as generative tools in design processes — inspiring ideation, highlighting overlooked attack surfaces, or prompting reconsideration of design assumptions before technologies are implemented.

We therefore position the vignette as an appropriate methodological form for a Provocations contribution: a lightweight, imaginable, and scalable medium that enables speculative artefacts to be encountered without technological implementation while remaining grounded in established HCI traditions, including the use of vignettes in survey research, design fiction, and scenario-based studies. By embedding each design within a relatable narrative situation, the vignettes activate readers' existing mental models and experiential knowledge, supporting reflective engagement with security and privacy risks that are otherwise difficult to visualize or experience directly.

We will provide some of our *stems*[3] for one example: Eleventh Finger. We felt that many of our designs were privacy-focused and we wanted a design that was explicitly security-focused. STRIDE is a threat modelling framework that outlines six high level categories of threats: Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, and Elevation of Privilege [27]. We considered possible designs for each and Eleventh Finger came from our ideation around Spoofing. We tried to avoid privacy implications, even though they are important as well (the ACLU and EPIC were raising concerns about Disney's use of biometric fingerprint scanning at its parks). Biometrics are often considered the holy grail of authentication by non-experts for their simplicity, however they have drawbacks in the areas of entropy, revocation after theft, and delegation. At the hacker conference Def Con, researchers showed that biometric scanners, like those found in modern smartphones, can be tricked using 3D-printed fingerprints [17]. We were open to using (dark) humour to avoid sounding too earnest in our designs. The reader can likely see how Eleventh Finger can emerge from these stems, however it is less apparent where the stems themselves came from (*e.g.,* from our academic backgrounds and personal experiences), nor the large number of stems we discarded because we could not see connections between them.

2.4 Future Work

In this Provocations paper, we presented the initial ideas for creating critical design vignettes for privacy and security concepts. Building on our reflections, we will explore a more systematic approach to the design creation process. We plan to first investigate an initial step in our methodology for brainstorming ideas for the designs using ideation cards. We will try different design methodologies from the literature (like exploring convergent and divergent methods, mapping and prototyping), and establish a reproducible methodology. Drawing on our initial experiments with generating

[3]In this context, 'stems' refer to the core ideas, influences, or conceptual elements that contribute to the development of a design.

designs, we aim to address the difficulties we encountered during this process (*i.e.,* coming up with topics related to security and privacy, attaining a level of plausibility etc.). By choosing the most relevant method from user research, we will also evaluate the effectiveness of our future methodology. We will also discuss whether our methodology can be applied to other areas in computer science like AI, robotics etc.

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